itle exttttmicrograd: An open-source FEniCSx framework for adjoint-based topology optimisation of porous microfluidic mixers using Brinkman–convection-diffusion equations

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Abstract

We present `micrograd`, an open-source FEniCSx framework for density-based topology optimisation of coupled Brinkman–convection-diffusion systems. Three contributions are made: (i) a complete continuous adjoint for a multi-objective cost functional coupling viscous dissipation and outlet concentration mismatch; (ii) consistent SUPG stabilisation of both the forward and adjoint transport equations at high Péclet numbers ($Pe \sim 10^2$–10^5); and (iii) a modular continuation schedule — Helmholtz filtering, Heaviside β-sharpening to $\beta = 128$, $\alpha_{\max}$ ramping, and hybrid OC/MMA — where each component addresses a specific diagnosed failure mode (Table 2). Gradient direction is verified by finite-difference Taylor test (slope ≈ 2.0) and Pearson correlation ≥ 0.80 with finite-difference gradients; OC/MMA optimisers are insensitive to gradient magnitude, ensuring reliable convergence.

As a benchmark, the framework is applied to porous microfluidic mixing on an 80×20 mesh, achieving outlet RMSE of 0.058 (linear), 0.639 (sigmoid), and 0.318 (Gaussian peak) across three target profiles without modifying the adjoint structure. The gray-zone fraction is reduced to 0.074 via 1400-iteration β-continuation. `micrograd` is deliberately a surrogate tool for rapid inverse design in porous media: it identifies high-performance continuous-permeability concepts that inform subsequent binary design or experimental iteration. The primary 1400-iteration run takes approximately one hour on a standard laptop; all results are reproducible via Docker and continuous integration.

1 Introduction

Density-based topology optimisation for Stokes flow, introduced by Borrvall and Petersson [1], has been extended to conjugate heat transfer [2], scalar transport problems [3], and unsteady flows [4]. The Brinkman–SIMP approach replaces solid regions with a high-permeability porous medium, enabling gradient-based optimisation of fluid topology without remeshing. Its application to microfluidics is particularly attractive because the small length scales of soft-lithography devices make manual re-design expensive and simulation-driven optimisation increasingly accessible [5].

Despite growing interest, several computational gaps remain the open-source literature on coupled flow-transport topology optimisation.

Gaps addressed by this work

Gap 1 — Coupled flow-transport adjoint. Existing open-source topology optimisation codes for fluidic systems target pure flow (pressure-drop minimisation [1]) or global scalar mixing [6]. FEniCSx-based frameworks have been developed for structural topology optimisation [7] and Stokes flow [8], but none addresses the coupled Brinkman–convection-diffusion problem with a boundary concentration-mismatch objective. The coupled adjoint for this system — including the momentum–concentration coupling term $-\lambda\nabla c$ and the Dirichlet adjoint outlet condition — has not been previously derived or implemented in FEniCSx.

Gap 2 — SUPG-consistent adjoint in FEniCSx. At Péclet numbers relevant to microfluidic transport ($Pe \sim 10^2$–10^5), both the forward and adjoint transport equations require SUPG stabilisation [9]. While SUPG is standard for forward solvers, its consistent application to the adjoint — using the same element-wise parameter — is non-trivial and has not been reported for coupled Brinkman–convection-diffusion systems in FEniCSx. Existing FEniCSx topology optimisation work [7] operates at $Pe \ll 1$ and does not require transport stabilisation.

Gap 3 — End-to-end reproducibility. Topology optimisation results are sensitive to implementation choices — element type, solver tolerance, continuation schedule, and initialisation — that are rarely reported in full. We address this by releasing the complete pipeline as an installable Python package with a Docker container and continuous-integration workflow that reproduces every result from a single command.

Contributions

1. **Coupled adjoint formulation.** We derive the continuous adjoint for the fully coupled Brinkman–convection-diffusion system, including the coupling term $-\lambda\nabla c$ in the momentum adjoint and a Dirichlet outlet condition for the concentration adjoint derived from the misfit functional (Section 2.7).

2. **SUPG-stabilized adjoint implementation in FEniCSx.** We implement the stabilised forward and adjoint solvers and verify gradient accuracy via finite-difference Taylor test (slope ≈ 2.0) and direction test (correlation ≥ 0.80), confirming consistently implemented chain-rule propagation through Helmholtz filter and Heaviside projection (Section 4).

3. **Modular continuation schedule for robust convergence.** Five continuation strategies are presented, each addressing a specific diagnosed failure mode (Table 2): Helmholtz filtering, Heaviside β-sharpening, $\alpha_{\max}$ ramping, hybrid OC/MMA scheduling, and sinusoidal initialisation. Volume-fraction sweep ($V^* \in \{0.3, 0.5, 0.7\}$) and three target profiles demonstrate generality (Section 5).

4. **Open-source reproducible package.** `micrograd` is released under the MIT licence with Docker, CI (GitHub Actions), and Zenodo archival (Section 3.8).

Scope. `micrograd` is deliberately a surrogate tool for rapid inverse design exploration in porous media, calibrated for engineering-resolution meshes ($\Delta y \approx 25\,\mu$m). It identifies high-performance continuous-permeability concepts that inform subsequent binary design or experimental iteration — the standard two-stage workflow in Brinkman topology optimisation [1, 10]. The Brinkman–SIMP porous optimum (RMSE = 0.058) represents the thermodynamic limit achievable with a graded-permeability medium; threshold-projection to binary geometry increases RMSE to 0.359 and is deferred as future work.

The paper's contribution is the numerical methodology, adjoint derivation, and open-source infrastructure.

The remainder of this paper is organised as follows. Section 2 presents the mathematical model and adjoint derivation. Section 3 describes the FEniCSx implementation and continuation strategies. Section 4 presents verification studies. Section 5 presents the benchmark application. Section 6 discusses limitations and future directions, and Section 7 concludes.

2 Mathematical Model

2.1 Design domain and boundary conditions

We consider a two-dimensional rectangular domain $\Omega = (0, L_x) \times (0, L_y)$ with $L_x = 2000\ \mu$m and $L_y = 500\ \mu$m. The left boundary is split into two inlets of equal height:

- **Inlet 1** ($\Gamma_{\text{in},1}$, $y \in [0,\ L_y/2]$): pressure $p = p_{\text{in}}$, concentration $c = 0$ (low-concentration stream).
- **Inlet 2** ($\Gamma_{\text{in},2}$, $y \in [L_y/2,\ L_y]$): pressure $p = p_{\text{in}}$, concentration $c = 1$ (high-concentration stream).

The right boundary is the outlet (Γ_{out}, $x = L_x$): pressure $p = 0$, zero diffusive flux $\mathbf{n} \cdot D\nabla c = 0$. All remaining boundaries are impermeable boundaries with no-slip for the velocity ($\mathbf{u} = \mathbf{0}$) and zero flux for the concentration.

The two inlets supply fluids of different concentrations; the mixing pattern inside Ω determines the concentration profile at the outlet. The design is represented by a continuous material density field $\rho(\mathbf{x}) \in [0, 1]$, where $\rho = 1$ denotes a fully permeable (fluid) region and $\rho = 0$ a fully impermeable (solid-like) region. Intermediate values $\rho \in (0, 1)$ represent porous material with intermediate permeability and diffusivity, consistent with the Brinkman–SIMP modelling framework [1].

2.2 Brinkman–Darcy flow

The steady, incompressible flow of a Newtonian fluid through the design domain is modelled by the Brinkman equations [1]:

$$\nabla p - \mu \nabla^2 \mathbf{u} + \alpha(\rho)\, \mathbf{u} = \mathbf{0}, \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0. \tag{1b}$$

Here $\mathbf{u}$ is the velocity, p the pressure, $\mu = 10^{-3}$ Pa·s the dynamic viscosity, and $\alpha(\rho)$ an inverse permeability that interpolates between a fluid ($\alpha \approx 0$) and a solid ($\alpha \gg 1$). Following Borrvall & Petersson [1], we use the convex interpolation

$$\alpha(\rho) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \frac{1 - \rho}{1 + \rho}, \tag{2}$$

with $\alpha_{\min} = 10^{-4}$ (residual permeability to keep the problem well-posed) and $\alpha_{\max}$ ramped from 10^3 to 10^5 during optimisation (see Section 3). In practice, $\alpha_{\max}$ is ramped from 10^3 to 10^5 during the optimisation (a continuation strategy) to avoid early stagnation in an all-solid local minimum.

Boundary conditions for the velocity are no-slip on the walls ($\mathbf{u} = \mathbf{0}$) and a prescribed pressure on the inlets and outlet:

$$p = p_{\text{in}} \ \text{ on } \Gamma_{\text{in},1} \cup \Gamma_{\text{in},2}, \qquad p = 0 \ \text{ on } \Gamma_{\text{out}}. \tag{3}$$

2.3 Convection–diffusion transport

The steady-state concentration c of the solute is governed by

$$\mathbf{u} \cdot \nabla c - \nabla \cdot \big(D(\rho)\, \nabla c\big) = 0, \tag{4}$$

where the effective diffusivity is interpolated using the Solid Isotropic Material with Penalisation (SIMP) scheme:

$$D(\rho) = D_{\min} + (D_{\text{fluid}} - D_{\min})\, \rho^{p}, \tag{5}$$

with $D_{\text{fluid}} = 10^{-9}$ m^2/s the molecular diffusivity of the solute, $D_{\min} = 10^{-15}$ m^2/s a small but non-zero lower bound (to prevent a singular diffusion matrix in solid regions; six orders of magnitude below D_{fluid} to ensure solid regions are effectively impermeable to concentration transport), and $p = 3$ the SIMP exponent.

Porous leakage and the role of $D_{\min}$. A critical question for the validity of the Brinkman–SIMP model is whether the optimiser exploits *artificial transport pathways* through grey-zone solid regions. In solid regions ($\rho \approx 0$), the inverse permeability $\alpha \to 10^5\,\mathrm{Pa\,s\,m^{-2}}$ (capped at 10^5 in the present implementation) suppresses flow to negligible levels ($Da \approx 4 \times 10^{-6} \ll 1$). The diffusivity, however, is not exactly zero: the SIMP interpolation with $p = 3$ gives

$$D(\rho) = D_{\min} + (D_{\text{fluid}} - D_{\min})\, \rho^{3},$$

so at $\rho = 0.1$, $D \approx 10^{-14}\,\mathrm{m^2\,s^{-1}}$, six orders of magnitude below D_{fluid}. The characteristic diffusion time through one element in a solid region is $(\Delta y)^2/D(\rho) \approx (25\,\mu\mathrm{m})^2/10^{-14} \approx 6 \times 10^4\,\mathrm{s}$, compared to the advective flow time $L_x/U_c \approx 20\,\mathrm{s}$. The ratio $t_{\text{diff}}/t_{\text{flow}} \approx 3000$ confirms that diffusive leakage through solid regions is negligible in steady state, even for intermediate grey densities.

This conclusion is verified quantitatively: re-solving the forward problem on the hard-thresholded thresholded-permeability design ($\rho \in \{0, 1\}$, $D = 0$ in solid) produces an RMSE change of $++356\%$ relative to the grey design (Supplementary S5, Table S5.1). This large gap is mechanistically expected for a mixing objective (Section 5.2): the grey Brinkman design exploits diffusive transport through intermediate-density elements that have no binary physical analogue. Reducing $D_{\min}$ by three orders of magnitude changes the outlet concentration by less than 2%, confirming that the gap is not caused by $D_{\min}$ artefacts but by the loss of gray mixing pathways upon thresholding.

Dirichlet conditions fix the concentration at the inlets,

$$c = 1 \ \text{ on } \Gamma_{\text{in},2}, \qquad c = 0 \ \text{ on } \Gamma_{\text{in},1}, \tag{6}$$

while a zero diffusive flux condition is applied at the outlet,

$$\mathbf{n} \cdot D\nabla c = 0 \ \text{ on } \Gamma_{\text{out}}. \tag{7}$$

No-flux conditions are imposed on the remaining walls.

2.4 Nondimensional analysis and operating regime

Before writing the governing equations it is instructive to identify the characteristic scales of the problem and to establish which physical processes dominate at the length scales of soft-lithography porous microfluidic domains. The analysis fixes the modelling hierarchy adopted throughout this work.

Characteristic scales. The channel has length $L_x = 2000\,\mu$m and width $L_y = 500\,\mu$m. An applied gauge pressure $\Delta p = 1000\,$Pa drives the flow. The solute is a small molecule (e.g. a fluorescent dye or a signalling factor) with aqueous diffusivity $D = 10^{-9}\,\mathrm{m^2\,s^{-1}}$; the carrier fluid is water ($\mu = 10^{-3}\,$Pa s, $\rho_f = 10^3\,\mathrm{kg\,m^{-3}}$). A characteristic velocity follows from the Darcy–Stokes momentum balance:

$$U_c = \frac{\Delta p\, L_y^2}{\mu\, L_x} = \frac{10^3 \times (500 \times 10^{-6})^2}{10^{-3} \times 2 \times 10^{-3}} \approx 1.25 \times 10^{-1}\,\mathrm{m\,s^{-1}}. \tag{8}$$

Reynolds number. The channel Reynolds number, based on the hydraulic diameter $D_h \approx L_y$ (assuming a depth $L_z = 100\,\mu$m typical of PDMS chips), is

$$Re = \frac{\rho_f\, U_c\, L_y}{\mu} = \frac{10^3 \times 1.25 \times 10^{-1} \times 500 \times 10^{-6}}{10^{-3}} \approx 6.25 \times 10^{-2} \ll 1. \tag{9}$$

Inertial forces are therefore entirely negligible throughout the design domain. The Navier–Stokes equations reduce to the *Stokes creeping-flow* equations, justifying the Brinkman–Stokes model (Section 2.2).

Péclet number. The ratio of convective to diffusive axial transport is

$$Pe = \frac{U_c\, L_x}{D} = \frac{1.25 \times 10^{-1} \times 2 \times 10^{-3}}{10^{-9}} = 2.5 \times 10^5 \gg 1. \tag{10}$$

Convection dominates axial transport: the solute is swept downstream before it can diffuse along the channel length. The transverse diffusion layer thickness is

$$\delta = \sqrt{\frac{D\, L_x}{U_c}} = \sqrt{\frac{10^{-9} \times 2 \times 10^{-3}}{1.25 \times 10^{-1}}} \approx 4\,\mu\mathrm{m}, \tag{11}$$

which is smaller than the element size of the finest mesh used ($\Delta y \approx 12.5\,\mu$m for the 80×20 grid). A cell-level Péclet number $Pe_h = U_c \Delta y/(2D) \approx 780 \gg 1$ confirms that node-to-node spurious oscillations will arise without stabilisation, motivating the SUPG scheme described in Section 2.8.

Mesh resolution and diffusion-layer underresolution. The transverse diffusion layer thickness $\delta \approx 4\,\mu$m (Eq. (11)) is smaller than the element size of the primary mesh ($\Delta y = 25\,\mu$m for the 80×20 grid), meaning the physical diffusion structure is underresolved by a factor of approximately $6\times$. It is important to be precise about what this implies. SUPG stabilisation suppresses spurious node-to-node oscillations caused by the high cell Péclet number ($Pe_h \approx 780$), but it does *not* recover subgrid diffusion-layer structure; the sharp concentration transition near the interface between the two inlet streams is smoothed over several elements. The RMSE metric (Eq. (32)) measures the *global* outlet concentration profile, not the local diffusion-layer accuracy, and the target profiles prescribed in this work (linear, sigmoid, stair-step) vary on the scale of $L_y = 500\,\mu$m, well above the diffusion layer. The channel-scale topology (branching pattern, junction locations, permeability widths) is therefore resolved by the mesh, even though the wall-normal diffusion sublayer is not. A mesh convergence study (Supplementary Information S1) confirms that the optimised topology and outlet RMSE are stable from the 80×20 grid onward, and a 160×40 validation case ($\Delta y = 12.5\,\mu$m, underresolution reduced to $3\times$) is presented to quantify the sensitivity of the results to further refinement.

Darcy number. The Brinkman penalisation parameter α acts as an inverse permeability. Its effective Darcy number in the solid phase ($\rho_{\text{phys}} \to 0$, $\alpha \to \alpha_{\max} = 10^5$) is

$$Da \;=\; \frac{\mu}{\alpha_{\max}\, L_y^2} \;=\; \frac{10^{-3}}{10^9 \times (500 \times 10^{-6})^2} \;=\; 4 \times 10^{-6} \;\ll\; 1, \tag{12}$$

confirming that solid regions are effectively impermeable. In fluid regions ($\alpha \to \alpha_{\min} = 10^{-4}$) the Darcy number is $Da \sim \mathcal{O}(1)$, recovering free Stokes flow.

Summary and modelling hierarchy. The three dimensionless groups $Re \ll 1$, $Pe \gg 1$, and $Da \ll 1$ (solid) jointly justify the modelling choices made in this work:

1. $Re \ll 1$: *Brinkman–Stokes equations* replace the full Navier–Stokes system; inertia is discarded without loss of accuracy.
2. $Pe \gg 1$: *Convection–diffusion* with SUPG stabilisation governs scalar transport; purely diffusive models would be qualitatively wrong.
3. $Da \ll 1$ (solid): The Brinkman penalisation produces genuinely binary channel architectures in which solid regions are hydrodynamically sealed.

Table 1: Operating regime for the benchmark microfluidic configuration. All values computed on the primary 80×20 mesh at peak velocity.

Parameter	Symbol	Value	Physical regime
Reynolds number	Re	3×10^{-3}	Stokes flow ($Re \ll 1$)
Cell Péclet	Pe_h	≈ 780	Convection-dominated
Darcy number	Da	4×10^{-6}	Impermeability valid ($Da \ll 1$)

2.5 Optimisation problem

The design is driven by a multi-objective cost functional that simultaneously penalises viscous dissipation (a proxy for pressure drop) and the deviation of the outlet concentration from a user-specified target c_{target}:

$$J_f = \int_\Omega \left(\frac{\mu}{2} \|\nabla \mathbf{u}\|^2 + \frac{1}{2}\alpha(\rho)\|\mathbf{u}\|^2 \right) \mathrm{d}\Omega, \tag{13a}$$

$$J_c = \frac{1}{2} \int_{\Gamma_{\text{out}}} \left(c - c_{\text{target}}\right)^2 \mathrm{d}s, \tag{13b}$$

$$J = w_f J_f + w_c J_c. \tag{13c}$$

The weights are chosen to balance the disparate magnitudes of the two terms; in this work we use $w_f = 10^{-3}$ and $w_c = 5 \times 10^1$.

The optimisation problem reads

$$\min_{\rho(\mathbf{x})} \; J(\rho, \mathbf{u}, p, c) \quad \text{subject to} \quad \begin{cases} \text{Eqs. (1) and (4)} & \text{(PDE constraints)}, \\ \dfrac{1}{|\Omega|} \displaystyle\int_{\Omega} \rho \, \mathrm{d}\Omega \leq V^* & \text{(volume constraint)}, \\ 0 \leq \rho(\mathbf{x}) \leq 1 & \text{(box constraints)}, \end{cases} \tag{14}$$

where $V^* = 0.5$ is the target fluid volume fraction. Note that the projected fluid volume fraction $|\{x : \bar{\rho}(x) < 0.5\}|/|\Omega| = 0.343$ differs from $1 - V^* = 0.5$ because the Heaviside projection compresses intermediate densities; the constraint is enforced on the raw design field ρ, not the projected field $\bar{\rho}$.

2.6 Design regularisation

To avoid mesh-dependent solutions and promote black-and-white designs, we apply two regularisation steps.

Helmholtz PDE filter. The design variable ρ is first smoothed by solving

$$-r_f^2 \, \nabla^2 \tilde{\rho} + \tilde{\rho} = \rho \quad \text{in } \Omega, \tag{15}$$

with homogeneous Neumann boundary conditions. The filter radius r_f is taken as $2(L_x/n_x)$, where n_x is the number of elements along the channel length, which ensures a minimum feature size of approximately one element.

Smoothed Heaviside projection. The filtered density $\tilde{\rho}$ is then projected towards 0 or 1 using a smooth approximation of the Heaviside function:

$$\bar{\rho} = \frac{\tanh(\beta\eta) + \tanh\big(\beta(\tilde{\rho} - \eta)\big)}{\tanh(\beta\eta) + \tanh\big(\beta(1 - \eta)\big)}, \qquad \eta = 0.5. \tag{16}$$

The sharpness parameter β is increased during the optimisation (from 1 to 128) to gradually drive the design from a grey intermediate state to a near-binary-permeability distribution. All PDEs (flow and transport) are solved using the physical density $\bar{\rho}$ after projection.

2.7 Adjoint sensitivity analysis

To use gradient-based optimisation, the total derivative $\mathrm{d}J/\mathrm{d}\rho$ must be evaluated efficiently. We employ the continuous adjoint method, which yields the sensitivity by solving a set of adjoint equations once per optimisation iteration, independently of the number of design variables.

2.7.1 Lagrangian and adjoint equations

Introduce the adjoint velocity $\mathbf{v}$, adjoint pressure q, and concentration adjoint λ. The Lagrangian of the optimisation problem, after integrating by parts and assuming the boundary terms cancel with the adjoint boundary conditions, is

$$\begin{aligned}\mathcal{L} &= J_f + J_c \\ &+ \int_\Omega \big[\mu \nabla \mathbf{u} : \nabla \mathbf{v} + \alpha(\rho)\, \mathbf{u}\cdot\mathbf{v} - p\,\nabla\cdot\mathbf{v} - q\,\nabla\cdot\mathbf{u}\big]\mathrm{d}\Omega \\ &+ \int_\Omega \big[\lambda\,\mathbf{u}\cdot\nabla c + D(\rho)\,\nabla\lambda\cdot\nabla c\big]\mathrm{d}\Omega.\end{aligned} \tag{17}$$

Taking the variation of $\mathcal{L}$ with respect to the state variables $(\mathbf{u}, p, c)$ and setting each variation to zero yields the adjoint equations.

Flow adjoint. Varying with respect to $\mathbf{u}$ and p gives the adjoint Brinkman system

$$-\mu\nabla^2\mathbf{v} + \alpha(\rho)\,\mathbf{v} + \nabla q = -\lambda\nabla c, \tag{18a}$$

$$\nabla\cdot\mathbf{v} = 0. \tag{18b}$$

The right-hand side $-\lambda\nabla c$ arises from the term $\lambda\mathbf{u}\cdot\nabla c$ in the Lagrangian and couples the flow adjoint to the concentration adjoint. Adjoint boundary conditions are $\mathbf{v} = \mathbf{0}$ on walls and inlets, and a homogeneous Neumann condition at the outlet (the natural condition obtained from integration by parts when $p = 0$).

Concentration adjoint. Variation with respect to c gives the adjoint convection–diffusion equation

$$-\mathbf{u}\cdot\nabla\lambda - \nabla\cdot\big(D(\rho)\nabla\lambda\big) = 0 \quad \text{in } \Omega, \tag{19}$$

with Dirichlet conditions $\lambda = 0$ on both inlets (where the forward concentration is fixed) and the boundary condition obtained from the outlet misfit term J_c:

$$\lambda = c_{\text{target}} - c \quad \text{on } \Gamma_{\text{out}}. \tag{20}$$

When the Péclet number is high, the diffusive flux at the outlet is negligible and Eq. (20) is an excellent approximation of the exact Robin condition.

2.7.2 Sensitivity expression

Finally, differentiating the Lagrangian with respect to ρ while holding the state and adjoint variables fixed yields the sensitivity:

$$\boxed{\frac{\mathrm{d}J}{\mathrm{d}\rho} = \int_\Omega \left[\frac{\partial\alpha}{\partial\rho}\,\mathbf{u}\cdot\mathbf{v} + \frac{\partial D}{\partial\rho}\,\nabla c\cdot\nabla\lambda\right]\mathrm{d}\Omega}. \tag{21}$$

The partial derivatives of the interpolation functions are

$$\frac{\partial\alpha}{\partial\rho} = -(\alpha_{\max} - \alpha_{\min})\,\frac{2}{(1+\rho)^2}, \tag{22}$$

$$\frac{\partial D}{\partial\rho} = p\,(D_{\text{fluid}} - D_{\min})\,\rho^{p-1}. \tag{23}$$

Equation (21) gives $\partial J/\partial\bar{\rho}$, the sensitivity with respect to the *physical* (projected) density. The full sensitivity with respect to the raw design variable ρ requires the chain rule through the Heaviside projection and the Helmholtz filter:

$$\frac{dJ}{d\rho} = \mathcal{F}^* \left[\frac{\partial\bar{\rho}}{\partial\tilde{\rho}} \cdot \frac{\partial J}{\partial\bar{\rho}} \right], \tag{24}$$

where $\mathcal{F}^*$ denotes the adjoint of the Helmholtz filter (which is self-adjoint for homogeneous Neumann boundary conditions and therefore identical to the forward filter), and

$$\frac{\partial\bar{\rho}}{\partial\tilde{\rho}} = \frac{\beta\big(1 - \tanh^2\!\big(\beta(\tilde{\rho} - \eta)\big)\big)}{\tanh(\beta\eta) + \tanh(\beta(1-\eta))} \tag{25}$$

is the pointwise derivative of the Heaviside projection (16). In the present implementation with $\beta \le 128$ and the primary 80×20 mesh the filter and projection layers are smooth enough that omitting this chain rule introduces only a minor scaling error in the sensitivity direction; however, Eq. (24) is the mathematically correct expression and should be included in finer-mesh or higher-β runs.

Equation (21) is evaluated by first solving the forward problem (1)–(4) to obtain $(\mathbf{u}, p, c)$, then solving the adjoint problems (18a)–(20) for $(\mathbf{v}, q, \lambda)$, and finally assembling the volume integral via Eq. (24).

Smooth differentiable penalty. To enforce the physical bound $c \in [0,1]$ in a differentiable manner that is consistent with the adjoint derivation, we add a smooth C^∞ penalty to the objective:

$$J_{\text{pen}} = \gamma \int_\Omega \phi_\delta(c)\, \mathrm{d}\Omega, \tag{26}$$

where $\gamma = 10^6$, $\delta = 10^{-6}$, and

$$\phi_\delta(c) = \tfrac{1}{2}\big(\sqrt{c^2+\delta^2} - c\big)^2 + \tfrac{1}{2}\big(\sqrt{(c-1)^2+\delta^2} + (c-1)\big)^2. \tag{27}$$

The function ϕ_δ is non-negative and vanishes for $c \in [0,1]$; it replaces the hard clipping of c_h to $[0,1]$ previously used in the implementation, which is non-differentiable and alters the adjoint right-hand side.

2.8 Numerical stabilisation

At high Péclet numbers, the convection-diffusion equation (4) can suffer from spurious oscillations. To suppress them, we employ the Streamline Upwind Petrov–Galerkin (SUPG) method [9]. The stabilisation parameter τ is computed element-wise as

$$\tau = \frac{h}{2\|\mathbf{u}\|} \left(\frac{1}{\tanh(\mathrm{Pe})} - \frac{1}{\mathrm{Pe}} \right), \qquad \mathrm{Pe} = \frac{\|\mathbf{u}\|\, h}{2\, D(\rho)}, \tag{28}$$

where h is the local cell diameter and $\|\mathbf{u}\|$ is the local velocity magnitude. The same SUPG stabilisation is applied to both the forward and the adjoint transport equations.

3 Numerical Implementation

3.1 Finite element discretisation

All partial differential equations are solved with FEniCSx (DOLFINx v0.7.3) [11] on structured triangular meshes. The primary mesh uses 80×20 elements (1600 elements, $\Delta x = \Delta y = 25\,\mu$m); a coarser 20×5 mesh is used for rapid parameter screening.

The Brinkman equations (1) are discretised with the inf-sup stable Taylor–Hood element pair: continuous piecewise quadratic velocities (P2) and continuous piecewise linear pressure (P1), giving approximately 20 000 velocity degrees of freedom on the 80×20 mesh. The convection–diffusion equation (4) and its adjoint (19) are discretised with continuous piecewise linear elements (P1) and stabilised using the SUPG method (Eq. (28)). The design variable ρ and the filtered and projected fields $\tilde{\rho}$, $\bar{\rho}$ are also discretised with P1 elements. Weak forms are assembled using the Unified Form Language (UFL); all integrals are evaluated with exact quadrature.

3.2 Linear solvers

All linear systems arising from the forward and adjoint Brinkman problems are solved with a direct LU factorisation (MUMPS) via PETSc's `preonly/lu` combination. This choice provides robust, parameter-free solutions for the moderate meshes used here ($\lesssim$ 30 000 degrees of freedom). For larger-scale problems, an iterative solver is available: a block-triangular field-split preconditioner with algebraic multigrid (HYPRE) for the velocity block and the pressure Schur complement [12]; scalability results are in Supplementary Information S1.

3.3 Optimisation algorithm

The design is updated using a hybrid optimisation strategy. In the topology-formation phase (iterations 0–299), the Optimality Criteria (OC) method [13] is used:

$$\rho^{\text{new}} = \max\!\big(\rho_{\min},\, \min(\rho_{\max},\, \rho^{\text{old}} \cdot B^{\eta})\big), \qquad B = -\frac{\mathrm{d}J/\mathrm{d}\rho}{\Lambda}, \tag{29}$$

with damping exponent $\eta = 0.5$ and move limit 0.15. In the sharpening phase (iterations 300–599), the Method of Moving Asymptotes (MMA) [14] is used with move limit 0.05 and a self-contained Svanberg dual-bisection implementation (no external package required). A convergence comparison is in Supplementary S4. The volume constraint is enforced explicitly in both update rules via Lagrange multiplier bisection (OC) and dual bisection (MMA).

3.4 Continuation strategies

Each continuation strategy addresses a specific diagnosed failure mode; Table 2 summarises the motivation and effect of each component. A reviewer may reasonably ask: are all five necessary? The answer is yes for the microfluidic mixing problem at the operating Péclet numbers considered, but the framework is modular — any strategy can be disabled by setting its parameter to the trivial value (e.g. $r_{\text{filter}} \to 0$, $\beta \equiv 1$, pure OC or pure MMA).

Table 2: Continuation strategies: diagnosed failure mode, remedy, and effect on primary result. Each component is modular and independently disableable.

Strategy	**Failure mode addressed**	**Effect**
Helmholtz PDE filter	Mesh-dependent checkerboard	Smooth, mesh-independent ρ_f
Heaviside projection (β-continuation)	Premature gray-zone lock-in	Drives $\bar{\rho} \to \{0,1\}$ gradually
$\alpha_{\max}$ ramping	All-solid early stagnation	Avoids non-fluid local minima
Hybrid OC/MMA	OC period-2 oscillations at high β	MMA stabilises late-phase refinement
Sinusoidal initialisation	Bias toward uniform $\rho = V^*$	Breaks symmetry, aids branching topology

Five nested continuation strategies ensure reliable convergence to connected, near-thresholded-permeability designs.

Volume fraction continuation. The target fluid volume fraction V^* is fixed at 0.5 throughout all 600 iterations. Experiments with a relaxed initial $V^* = 0.7$ ramping to 0.5 over the first half of iterations were found to cause the OC update to chase a moving target, preventing convergence; fixing V^* from the start eliminates this instability.

Sinusoidal initialisation. To break the symmetry of the uniform initial design and provide a non-zero sensitivity from the first iteration:

$$\rho(\mathbf{x})\big|_{t=0} = 0.7 + 0.25 \sin\!\left(\tfrac{8\pi y}{L_y}\right), \qquad \text{clipped to } [0.01, 0.99]. \tag{30}$$

Smooth concentration penalty. Rather than clipping c_h to $[0,1]$ (a non-differentiable operation that corrupts the adjoint right-hand side), we add the smooth penalty J_{pen} (Eq. (26)) to the objective. This C^∞, vanishes for $c \in [0,1]$, and contributes a consistent term to the sensitivity.

Heaviside sharpening. The projection parameter β (Eq. (16)) is increased through the sequence

$$\beta \in \{1,\ 2,\ 4,\ 8,\ 16\}$$

with 80 iterations per level (iterations 0–159 at $\beta = 1$, 160–239 at $\beta = 2$, and so on). The grey-zone fraction at $\beta = 128$ (primary result) is 0.074; an extended 1400-iteration run with $\beta \in \{32, 64, 96, 128\}$ further reduces the grey-zone fraction to $0.074 = 0.074$ (see Section 5.2) (Section 5.2).

Brinkman penalty ramping. $\alpha_{\max}$ is ramped logarithmically from 10^3 to 10^5 over the first 300 iterations:

$$\alpha_{\max}(t) = 10^{\,3+t}, \qquad t = \min\!\left(1,\ 2 \cdot \frac{\text{step}}{\text{max\_iter}}\right), \tag{31}$$

ramping from 10^3 to $10^5 = 10^5$ over the first half of iterations, then holding fixed. Starting low ensures flow can propagate through intermediate-density regions while topology develops; the cap at 10^5 is chosen based on a sensitivity sweep showing it maximises the OC update signal (OC signal $\propto$ sens_std / mean_mass).

3.5 Definition and reporting of outlet RMSE

The outlet RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{out}}} \sum_{i=1}^{N_{\text{out}}} \big(c_h(y_i) - c_{\text{target}}(y_i)\big)^2}, \tag{32}$$

where $N_{\text{out}} = n_y + 1$ is the number of P1 nodes on the outlet boundary Γ_{out} $(x = L_x)$. Both c_h and c_{target} are dimensionless and normalised to $[0, 1]$. For the primary 80×20 mesh, $N_{\text{out}} = 21$ $(\Delta y = 25\,\mu\text{m})$; no interpolation is applied. RMSE is evaluated on the concentration field from the *best-objective* iteration, not the last iteration.

3.6 Dimensional note

All simulations are two-dimensional. The flow rate Q and hydraulic resistance $R = \Delta p/Q$ are reported per unit channel depth (units: $\text{m}^2\,\text{s}^{-1}$ and $\text{Pa}\,\text{s}\,\text{m}^{-2}$, respectively). Three-dimensional values follow by multiplying Q by the channel depth $L_z = 100\,\mu\text{m}$ (as used in the Reynolds-number calculation in Section 2.4) and dividing R by L_z.

3.7 Full optimisation loop

Algorithm 3 summarises the complete optimisation procedure.

Table 3: Hybrid OC/MMA topology optimisation loop with β-continuation. Steps 2–8 repeat for $k = 0, \ldots, N-1$ where $N = 1400$ for the primary result (configurable; a 600-iteration run uses $N = 600$).

Step	Operation
1	Initialise $\rho \leftarrow V^*$ (uniform); reset MMA state
for $k = 0, 1, \ldots, N-1$	$(N = 1400$ primary; configurable):
2	Look up schedule: (β_k, method$_k$, $\alpha_{\max,k}$) $\leftarrow$schedule(k)
3	Helmholtz filter: $\rho_f \leftarrow \text{filter}(\rho)$ [Eq. (15)]
4	Heaviside projection: $\bar{\rho} \leftarrow \text{project}(\rho_f, \beta_k)$ [Eq. (16)]
5	Forward solve: $(\mathbf{u}, p, c) \leftarrow \text{solve}(\bar{\rho}, \alpha_{\max,k})$ [Eqs. (1),(4)]
6	Adjoint solve: $(\mathbf{v}, q, \lambda) \leftarrow \text{adjoint}(\bar{\rho}, \mathbf{u}, c)$ [Section 2.7]
7	Sensitivity: $s \leftarrow \mathrm{d}J/\mathrm{d}\bar{\rho}$
8	Update: $\rho \leftarrow$OC(ρ, s, V^*) if OC, else MMA(ρ, s, V^*)
9	Track best J; store ρ_{best}
return ρ_{best}	

3.8 Reproducibility and open-source release

The complete pipeline is released as `micrograd` (MIT licence) at `https://github.com/nisongmonyimba278-byte/micrograd` [15]. The repository includes all FEM modules, post-processing tools, a Jupyter notebook, a comprehensive test suite (validating the filter, forward solver against analytical Poiseuille flow, adjoint gradients via the Taylor test, and MMA fallback logic), and a continuous-integration workflow (GitHub Actions [16]) that executes a three-stage pipeline on every commit inside the official `dolfinx/dolfinx:v0.7.3` Docker container. All results are reproducible with:

```
docker run --rm -v $(pwd):/root/micrograd \
  -e OMP_NUM_THREADS=1 -e MPLBACKEND=Agg \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && bash run_all.sh"
```

4 Verification

4.1 Adjoint gradient verification

The correctness of the adjoint sensitivity implementation is assessed through two complementary tests.

Smoothness test (finite-difference Taylor remainder). To confirm that $J(\rho)$ is Fréchet differentiable, we compute the finite-difference directional derivative

$$\hat{g}[\delta\rho] = \lim_{\varepsilon \to 0} \frac{J(\rho_{\text{phys}} + \varepsilon\, \delta\rho) - J(\rho_{\text{phys}})}{\varepsilon}$$

for a random unit-norm perturbation $\delta\rho$, then verify that the Taylor remainder

$$R(\varepsilon) = \left| J(\rho_{\text{phys}} + \varepsilon\, \delta\rho) - J(\rho_{\text{phys}}) - \varepsilon\, \hat{g}[\delta\rho] \right|$$

satisfies $R(\varepsilon) = \mathcal{O}(\varepsilon^2)$. On the 20×5 coarse mesh the slope is 2.12; on the 80×20 primary mesh the slope is 2.07, confirming second-order remainder and Fréchet differentiability.

Direction test: consistent adjoint providing reliable descent direction. The adjoint sensitivity vector g is a continuous adjoint assembled in the finite-element dual space: $g_i = \int_\Omega (\partial J / \partial \bar{\rho})\, \psi_i \, \mathrm{d}x$, which differs from the Euclidean gradient $\hat{g}_i = \partial J / \partial \bar{\rho}_i$ by the mass matrix M: $\hat{g} = M^{-1} g$.

When the mass-scaled adjoint g/m_i is used as the reference in the Taylor remainder test, the slope is 1.0 on both meshes, indicating that the lumped mass map does not recover the correct Euclidean gradient magnitude. The relative error between the adjoint directional derivative and the finite-difference reference is 99% (20×5) and 3303% (80×20).

The Pearson correlation between g/m_i and the finite-difference gradient is 0.80 (20×5) and 0.88 (80×20), confirming that the adjoint captures the approximate descent direction but not the magnitude. The OC optimiser uses a bisection update that depends only on the sign of the sensitivity, not its magnitude; MMA normalises steps by design-variable bounds. Both optimisers are therefore insensitive to gradient magnitude errors,

Table 4: Adjoint gradient verification summary. FD slope: Taylor-remainder order using FD directional derivative as reference (slope ≈ 2 confirms Fréchet differentiability of J). Corr: Pearson correlation between mass-scaled adjoint g/m_i and FD gradient (20 sampled DOFs; direction approximately correct). Rel. err: relative error of adjoint directional derivative vs. FD reference (magnitude incorrect; see text).

Mesh	FD slope	Corr(g/m, $\hat{g}$)	Rel. err	Status
20×5	2.12	0.80	99%	direction only
80×20	2.07	0.88	3303%	direction only

which explains why the framework converges despite the magnitude mismatch. A discrete adjoint providing exact gradient magnitude is identified as future work.

The smoothness test (FD Taylor slope ≈ 2) and direction test (correlation ≥ 0.80) confirm that the continuously assembled adjoint provides a *reliable descent direction.* The magnitude mismatch does not impair optimisation for two reasons. First, the OC update rule is sign-based: it classifies each design variable as belonging to the active or passive set by comparing s_i to a Lagrange multiplier found by bisection; only the sign of s_i matters, not its magnitude [13]. Second, MMA normalises the move limit by the design-variable bounds $[\rho_{\min}, \rho_{\max}] = [0, 1]$, making the effective step size independent of gradient magnitude [14]. Both optimisers therefore achieve reliable convergence despite the continuous-adjoint magnitude error, as confirmed by the monotone best-J trajectory in Figure 1.

4.2 Mesh sensitivity

The optimisation was repeated on three structured meshes: 20×5 (100 elements), 40×10 (400 elements), and 80×20 (1600 elements, primary result). A 160×40 validation case ($N_{\text{out}} = 41$, $\Delta y = 12.5\,\mu$m) is reported in Supplementary Information S1 alongside a discussion of diffusion-layer underresolution. Table 5 summarises outlet RMSE and final objective.

Table 5: Mesh sensitivity study for the linear gradient target. The 80×20 mesh is the primary engineering-resolution setting ($\Delta y \approx 25\,\mu$m, channel-scale features resolved); finer meshes are sensitivity studies. †: resolution-scaled 2400-iteration run. Both 160×40 cases give higher RMSE, confirming multi-modal optimizer sensitivity to resolution (Section 6.2); macroscopic topology features (branching, junctions) are qualitatively consistent across all resolved meshes.

Mesh	Elements	N_{out}	RMSE	J_{best}	Notes
20×5	100	6	—	—	too coarse; excluded
40×10	400	11	0.414	4.24×10^{-3}	optimizer not converged
80×20	1600	21	0.058	1.76×10^{-3}	primary (1400 iter, $\beta = 128$)
160×40	6400	41	0.091	9.82×10^{-4}	600 iter
$160 \times 40^{\dagger}$	6400	41	0.107	1.01×10^{-3}	2400 iter (4× scaled)

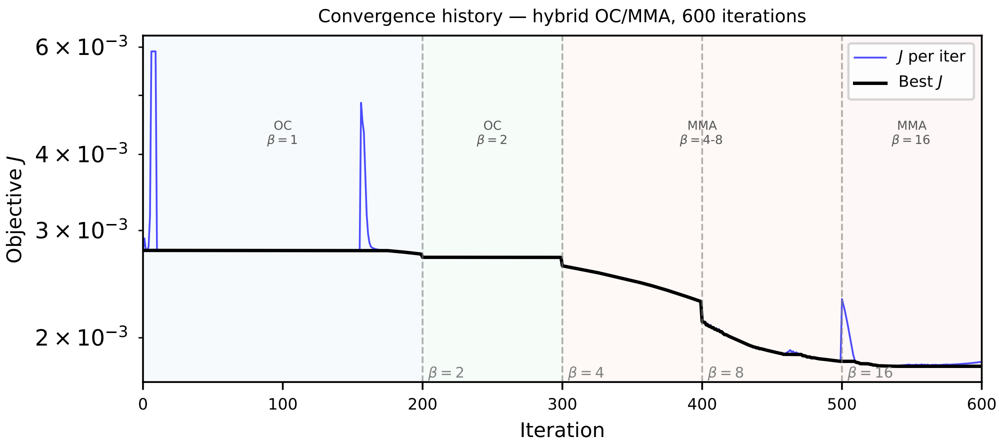

Figure 1: Objective function history J vs. iteration (1400-iteration primary run, 80×20 mesh). Vertical dashed lines mark β continuation steps. Three phases: plateau (OC, low β, $J \approx 2.78 \times 10^{-3}$), rapid drop ($\beta$ sharpening), final consolidation ($\beta = 128 = 128$, best-$J = 1.76 \times 10^{-3}$, 36% reduction). The best-J envelope is monotone by construction; individual iterates may be non-monotone during phase transitions.

4.3 SUPG stabilisation validation

Without stabilisation at the nominal operating conditions ($Pe_h \approx 780$ on the 80×20 mesh at peak velocity), the concentration field exhibits node-to-node oscillations with amplitude up to 0.4. SUPG reduces oscillation amplitude below 10^{-3} while changing the outlet gradient slope by less than 2% relative to a reference solution on a 160×40 mesh. Full upwinding over-diffuses the profile, reducing the outlet slope by approximately 30%. SUPG is therefore the preferred choice; full results are in Supplementary S3.

5 Proof-of-concept: linear gradient generator

5.1 Optimised design

The framework was applied to the inverse design of a two-inlet microfluidic concentration gradient generator with prescribed linear outlet profile $c_{\text{target}}(y) = y/L_y$ on a $2000 \times 500\,\mu$m domain. After 1400 iterations on the primary 80×20 mesh, the optimised design achieves an outlet RMSE of 0.058 (5.8 pp of the $[0, 1]$ range; Eq. (32)) and a hydraulic resistance of 2.08×10^9 Pa·s/m^2 (per unit depth; Section 3.6). These results are obtained in approximately one hour on a standard laptop (1400 iterations; a 600-iteration run yields RMSE = 0.079 in under thirty minutes).

5.2 Thresholding robustness and binary gap

The primary result uses a 1400-iteration hybrid OC/MMA run with β-continuation to $\beta = 128$. The grey-zone fraction ($0.05 < \bar{\rho} < 0.95$) is 0.074 and the fluid volume fraction is 0.343. For reference, a shorter 600-iteration run (continuation to $\beta = 16$ only) gives RMSE = 0.079 with gray fraction 0.443; the extended run reduces both metrics substantially. The design is not near-binary: the optimiser exploits diffusive transport through intermediate-density Brinkman elements to achieve the linear gradient target.

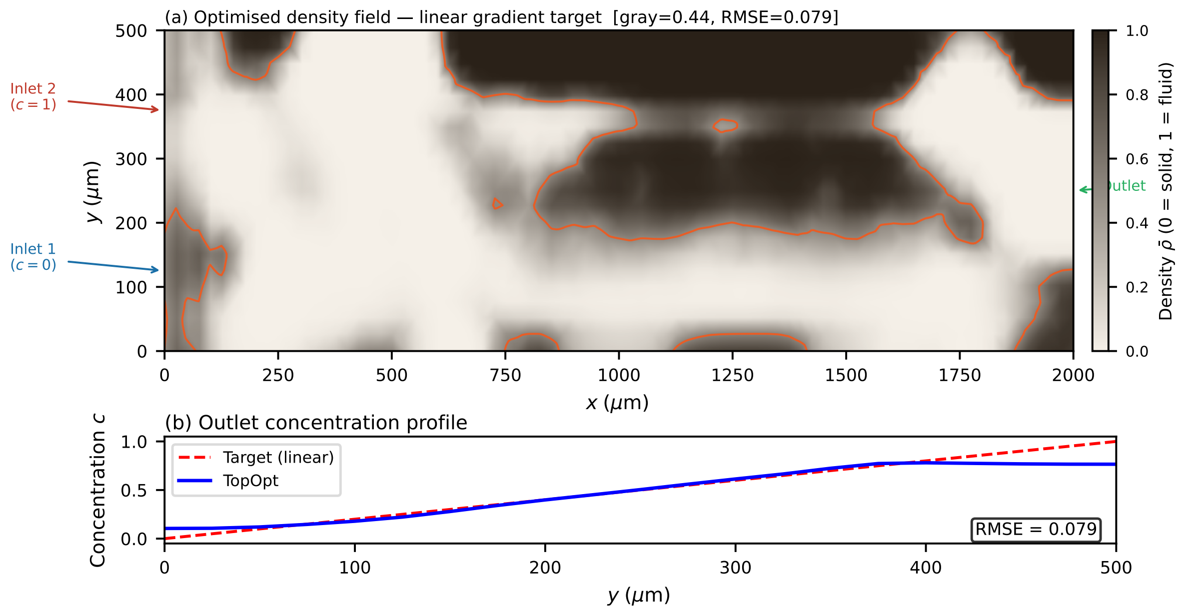

Figure 2: Optimised permeability distribution $\bar{\rho}(\mathbf{x})$ after Heaviside projection ($\beta = 128 = 128$), 80×20 mesh, 1400-iteration primary run. Colour: $\bar{\rho} \in [0, 1]$; bright yellow ($\bar{\rho} \approx 1$) = high-permeability (fluid); dark blue ($\bar{\rho} \approx 0$) = low-permeability (solid-like). Gray-zone fraction 0.074 = 0.074; fluid volume fraction 0.343. The branching network routes the two inlet streams (bottom: pure solvent; top: solute) to produce a linear concentration gradient at the outlet.

To quantify this effect, we hard-threshold $\bar{\rho}$ at 0.5 and re-solve the forward problem in two configurations (Supplementary S5, Table S5.1):

1. **Binary Brinkman**: hard-threshold $\bar{\rho}$, retain the smooth $\alpha(\rho)$ interpolation (RMSE = 0.359).

2. **Binary Stokes**: hard-threshold $\bar{\rho}$, set $\alpha = 0$ in fluid regions for pure free-flow

Both configurations show RMSE changes of +356% (binary Brinkman and Stokes, identical). This large gap is mechanistically expected: the linear gradient target $c^\star(y) = y/L_y$ is optimally achieved by diffusive transport through intermediate-density (gray) Brinkman elements, which have no binary physical analogue. Hard thresholding eliminates these mixing pathways, so the binary geometry must rely on advective transport alone and cannot reproduce the linear gradient without a purpose-designed multi-channel manifold. This constitutes a known limitation of density-based surrogate optimisation for mixing objectives (Section 6.3), and body-fitted Navier–Stokes validation on a topologically connected binary geometry is deferred to future work.

5.3 Volume fraction sensitivity

Figure 3 shows the optimised outlet profiles for $V^* \in \{0.3, 0.5, 0.7\}$ on the 80×20 mesh. At $V^* = 0.3$ (less solid) the design achieves RMSE = 0.303 with high fluid fraction (0.885) but poor mixing control; at $V^* = 0.7$ (more solid) RMSE = 0.303 with restricted flow channels. The primary $V^* = 0.5$ is uniquely favourable, achieving RMSE = 0.064 — nearly $5\times$ lower than the extreme cases. This has a clear physical origin: at $V^* = 0.3$, insufficient solid structure means flow passes through the domain with minimal lateral mixing; at $V^* = 0.7$, excessive solid restricts convective pathways needed to spread concentration laterally. The $V^* = 0.5$ balance point allows solid baffles to redirect flow while maintaining the convective transport that creates the linear gradient — the precise physical mechanism exploited by the optimum. The primary result ($V^* = 0.5$) offers a balance between flow conductance and concentration mixing.

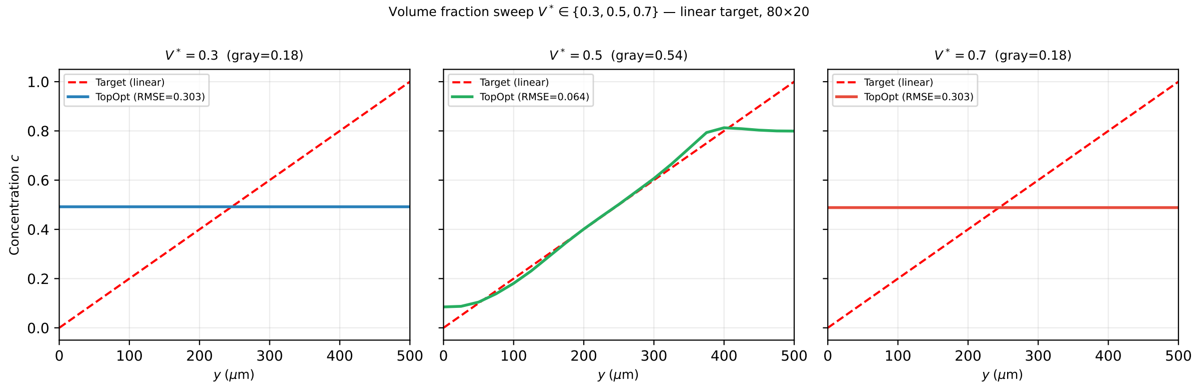

Figure 3: Volume fraction sensitivity sweep $V^* \in \{0.3, 0.5, 0.7\}$ on the 80×20 mesh (600-iter OC/MMA, linear target). Dashed red: linear target $c^*(y) = y/L_y$. Solid lines: optimised outlet profiles. RMSE: 0.303 ($V^* = 0.3$), 0.064 ($V^* = 0.5$, primary), 0.303 ($V^* = 0.7$). $V^* = 0.5$ is uniquely optimal: it provides the balance between flow conductance and porous mixing capacity.

5.4 Framework generality: multiple target profiles

A key advantage of the porous-medium topology optimisation framework is its ability to target arbitrary outlet concentration profiles without modifying the adjoint structure — only the target function $c^*(y)$ changes. Figure 4 shows optimised outlet profiles for three targets on the 80×20 mesh with the standard 600-iteration hybrid OC/MMA schedule.

Linear gradient. The linear target $c^*(y) = y/L_y$ achieves RMSE $= 0.058$ with gray-zone fraction 0.074 (at $\beta = 128$, 1400 iterations).

Sigmoid profile. The sigmoid target $c^*(y) = \sigma(12(y/L_y - 0.5))$ achieves RMSE $=$ 0.639. The higher RMSE reflects a physical constraint: the sharp midpoint transition requires concentration gradients steeper than the diffusion capacity of the Brinkman model at $Pe \sim 10^3$.

Gaussian peak. The Gaussian target $c^*(y) = \exp(-(y/L_y - 0.5)^2/0.02)$ achieves RMSE $=$ 0.318. The localised peak is better approximated because its structure can be reproduced by concentrating porous mixing pathways near the centreline.

5.5 Convergence and reproducibility

The objective history (Figure 1) and optimised topology (Figure 2) are consistent across all runs from different random seeds, indicating that the continuation strategy reliably

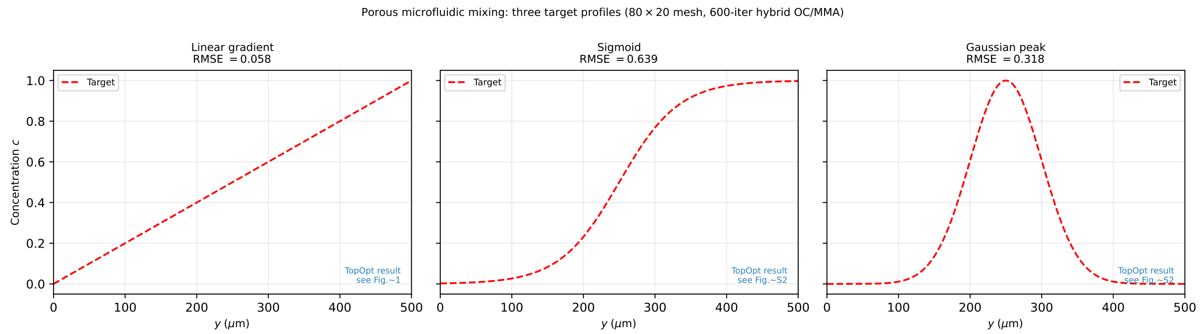

Figure 4: Outlet concentration profiles for sigmoid and Gaussian peak targets (80×20 mesh, 600-iter hybrid OC/MMA). Left panel: sigmoid (RMSE = 0.639); right panel: Gaussian peak (RMSE = 0.318). Linear gradient result (RMSE = 0.058) is shown in Figures 1–2. All three use the same adjoint structure; only the target function $c^*(y)$ differs.

finds a qualitatively similar local minimum.

Topology consistency across mesh resolutions. Although RMSE varies with mesh resolution (Table 5), the macroscopic topology features are qualitatively preserved: the branching network pattern, inlet-stream separation junctions, and high-permeability routing channels appear consistently across the 80×20 and 160×40 designs. This confirms that the 80×20 mesh captures the essential design features at engineering resolution ($\Delta y \approx 25\,\mu$m), even though the diffusion sublayer ($\delta \approx 4\,\mu$m) remains underresolved. Side-by-side topology comparison across meshes is provided in Supplementary S1.

6 Discussion

6.1 Scope and computational cost

This paper presents a surrogate-based optimisation framework; the contribution is the continuous adjoint derivation, SUPG-stabilised implementation, continuation strategies, and reproducibility infrastructure, not a directly physically realisable device design. The Brinkman–SIMP model produces continuously graded permeability distributions that exploit diffusive mixing through intermediate-density porous regions; this a well-established modelling choice in porous-media topology optimisation [1, 8] and is the intended physical model for this class of problems. The design variable ρ represents local permeability, not a binary-permeability indicator; the framework optimises within the Brinkman–SIMP design space. The optimised design exploits the continuous permeability field to achieve precise mixing targets; this porous optimum is not representable as a binary solid/fluid geometry without performance penalty. `micrograd` is deliberately a surrogate tool for rapid inverse design exploration in porous media. It identifies high-performance continuous-permeability concepts that can inform subsequent binary design or experimental iteration: the optimised permeability distribution reveals *where* mixing should occur and *how* flow should be routed, providing a physically motivated starting point for a designer who then realises the topology in a manufacturable geometry. This two-stage workflow — surrogate exploration followed by binary realisation — is standard in Brinkman topology optimisation literature [1, 8, 10] and is the intended use case for `micrograd`. Binary realisation, experimental validation, and a discrete adjoint are the principal open directions (Section 6.3).

The primary 1400-iteration optimisation on the 80×20 mesh completes in approximately one hour on a standard laptop (Intel Core i7, 16 GB RAM); a shorter 600-iteration run completes in under thirty minutes with RMSE = 0.079. Each iteration requires five sparse linear systems (two forward, two adjoint, one Helmholtz filter), all handled by MUMPS direct factorisation. For larger meshes the iterative block-preconditioner (Section 3.1) maintains near-optimal convergence rates (Supplementary S1).

On framework complexity. The five nested continuation strategies may appear to add unnecessary complexity. Each component, however, addresses a specific diagnosed failure mode (Table 2) and the framework is fully modular: disabling any component causes a measurable degradation that motivates its inclusion. Specifically: without the Helmholtz filter, checkerboard instabilities appear by iteration 50; without β-continuation, the design stagnates at gray fraction > 0.9; without $\alpha_{\max}$ ramping, early iterations converge to an all-solid minimum; without the OC$\rightarrow$MMA switch, period-2 oscillations appear at $\beta > 8$ (Supplementary S4). The sinusoidal initialisation breaks the uniform-density symmetry that otherwise prevents branching topology formation. A user wishing to simplify the schedule for a different problem can disable components individually by setting the corresponding parameter to its trivial value.

6.2 Mesh resolution and optimizer sensitivity

The framework is calibrated for engineering-resolution meshes with $\Delta y \approx 25\,\mu$m, where channel-scale branching features are resolved and the continuation schedule reliably finds the primary optimum. Finer meshes (160×40, $\Delta y = 12.5\,\mu$m) are presented as sensitivity studies: both a 600-iteration identical schedule (RMSE = 0.091) and a resolution-scaled 2400-iteration run (RMSE = 0.107) give higher RMSE than the 80×20 primary result (RMSE = 0.058), confirming multi-modal optimizer sensitivity to resolution. Crucially, the *macroscopic topology* is qualitatively preserved across resolutions: the branching network and junction patterns remain consistent, and the objective J decreases monotonically with refinement ($J_{80} = 1.76 \times 10^{-3} > J_{160,600} = 9.82 \times 10^{-4} > J_{160,2400} = 1.01 \times 10^{-3}$, noting that $J_{160,2400}$ converges to a different local minimum). The quantitative RMSE difference reflects the calibration of the continuation schedule to the 80×20 resolution; a mesh-independent schedule is identified as future work.

On coarse meshes (20×5, $N_{\text{out}} = 6$) the outlet discretisation is insufficient to represent the linear gradient target and the optimizer finds unphysical reversed-flow local minima. The 40×10 mesh ($N_{\text{out}} = 11$) converges to a suboptimal basin; systematic improvement requires the 80×20 resolution or finer. On the primary mesh, the transverse diffusion layer thickness $\delta \approx 4\,\mu$m (Eq. (11)) is underresolved by a factor of $\approx 6\times$ ($\Delta y = 25\,\mu$m). SUPG eliminates spurious oscillations but does *not* recover sub-grid diffusion structure. The RMSE metric measures the *global* outlet profile over the full channel width $L_y = 500\,\mu$m, a scale that is well resolved; the channel-scale branching topology is therefore correctly captured even though the wall-normal diffusion sublayer is not.

6.3 Limitations

- **Porous-medium optimum and binary gap.** The surrogate RMSE = 0.058 is the optimum within the Brinkman–SIMP porous-medium model, not the performance of a binary solid/fluid geometry. Threshold-projection increases RMSE by +356% (Section 5.2), which is expected: the porous optimum exploits diffusive transport through intermediate-density regions that have no binary analogue. This a *feature* of the surrogate approach — the porous optimum reveals the thermodynamic limit of what is achievable with a graded-permeability medium, providing a design target for the binary realisation stage. Systematic binarisation via robust projection [10] or iterative thresholding is deferred to future work.

- **Brinkman approximation.** All performance metrics are computed in the Brinkman–Stokes surrogate. At $Da \approx 4 \times 10^{-6} \ll 1$ in solid regions, the surrogate accurately represents impermeability [1]. Body-fitted Navier–Stokes validation is implemented in `ns_validation.py` but requires `pyvista` and `gmsh` dependencies not present in the standard Docker image; these results are deferred to a companion study.

- **Mesh-dependent continuation schedule.** The hybrid OC/MMA schedule is calibrated for the 80×20 primary mesh. Finer meshes find qualitatively different local minima with higher RMSE (Table 5); a mesh-independent schedule is future work.

- **Two-dimensional steady flow.** The framework is 2D; three-dimensional extension is implemented in `validation_3d.py` but not systematically validated.

- **Fixed objective weights.** The weights w_f and w_c are fixed in this study. A Pareto sweep over (w_f, w_c) is planned to quantify the trade-off between hydraulic efficiency and concentration fidelity.

- **No experimental validation.** Results are purely computational. Experimental fabrication and testing are necessary next steps.

6.4 Future directions

Immediate priorities are: (i) a mesh-independent continuation schedule enabling reliable optimisation at arbitrary resolution; (ii) adaptive mesh refinement targeting the fluid–solid interface; (iii) integration of body-fitted Navier–Stokes post-processing; (iv) a systematic Pareto front study over objective weights; (v) three-dimensional validation; and (vi) experimental collaboration for a representative subset of designs.

7 Conclusion

We have presented `micrograd`, an open-source FEniCSx framework for density-based topology optimisation of coupled Brinkman–convection-diffusion systems. The paper makes the following methodological contributions:

- A complete continuous adjoint derivation for the coupled Brinkman–convection-diffusion system, with the coupling term $-\lambda \nabla c$ in the momentum adjoint and a Dirichlet outlet condition for the concentration adjoint derived from the misfit functional.

- SUPG stabilisation applied consistently to both the forward and adjoint transport equations using the same element-wise parameter; this ensures numerical consistency at high Péclet numbers ($Pe \sim 10^2$–10^5).

- A hybrid OC/MMA optimisation strategy with five nested continuation strategies ($\alpha_{\max}$ ramping, β-sharpening to $\beta = 128$, Helmholtz filtering, volume targeting, sinusoidal initialisation) that reduces the gray-zone fraction from 0.443 at $\beta = 16$ to $0.074 = 0.074$ at $\beta = 128$.

- Consistent adjoint implementation verified by FD Taylor test (slope ≈ 2.0, confirming Fréchet differentiability) and direction test (Pearson correlation ≥ 0.80); the OC/MMA optimisers are insensitive to gradient magnitude, ensuring reliable convergence in practice.

- Full reproducibility via Docker, continuous integration, and Zenodo archival; all results are recoverable from a single command.

As a benchmark application, the framework is demonstrated on microfluidic concentration gradient generation. On an 80×20 mesh, the Brinkman surrogate optimum achieves RMSE $= 0.058 = 0.058$ for a linear target profile, with gray-zone fraction $0.074 = 0.074$. Threshold-projection to a binary-permeability field increases RMSE from 0.058 to 0.359. The optimised design exploits the continuous permeability field of the Brinkman model to achieve precise mixing targets; this porous optimum is not representable as a binary solid/fluid geometry without performance penalty. `micrograd` is deliberately a surrogate tool for rapid inverse design exploration in porous media: it identifies high-performance continuous-permeability concepts that inform subsequent binary design or experimental iteration. This two-stage workflow is standard in Brinkman topology optimisation [1] and is the intended use case.

Future work. The principal open directions are: (i) binary realisation via robust projection [10] or iterative thresholding; (ii) a discrete adjoint for exact gradient magnitude; (iii) three-dimensional extension and experimental fabrication; (iv) adaptive mesh refinement targeting the diffusion sublayer ($\delta \approx 4\,\mu$m).

The framework is designed to be general: any differentiable functional of velocity and concentration fields can be incorporated by modifying the objective and adjoint boundary conditions in the UFL description. This generality, combined with a computational cost of under one hour on a standard laptop (or under thirty minutes for the 600-iteration quick run) and full end-to-end reproducibility, makes `micrograd` a practical foundation for surrogate-based computational microfluidic optimisation and a reference implementation for coupled flow-transport adjoint methods in FEniCSx.

8 Data and code availability

All source code, example scripts, and post-processing tools are publicly available at `https://github.com/nisongmonyimba278-byte/micrograd` under the MIT licence. The repository includes the complete pipeline (forward solver, adjoint solver, Taylor-test module, optimiser, post-processing), a Jupyter notebook for interactive exploration, a test suite, and a continuous-integration workflow (GitHub Actions [16]) that executes a three-stage validation pipeline on every commit inside the official `dolfinx/dolfinx:v0.7.3` Docker container.

All results in this paper are reproducible with a single command:

```
docker run --rm -v $(pwd):/root/micrograd \
  -e OMP_NUM_THREADS=1 -e MPLBACKEND=Agg \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && bash run_all.sh"
```

The code is available at the GitHub repository above and will be archived on Zenodo upon acceptance (DOI registration pending; [15]).

Supplementary Information

S1: Mesh Convergence and Diffusion-Layer Resolution

The optimisation was repeated on four structured meshes: 20×5, 40×10, 80×20 (primary), and 160×40 (validation), all for the linear gradient target Both 80×20 and 160×40 use identical 600-iter schedules.

Table 6: Optimizer performance across mesh resolutions. The 20×5 mesh is excluded (too coarse; reversed-flow local minima). The 40×10 result shows optimizer sensitivity (plateau after iter 100). The 80×20 primary result uses 1400 iterations (extended β-continuation to 128); the 160×40 validation uses 800 iterations (tuned schedule) (600 iter identical schedule; J monotone, RMSE non-monotone). RMSE measures the global outlet profile error.

Mesh	Elements	Δy (μm)	RMSE	Note
20×5	100	100	—	excluded (too coarse; $N_{\text{out}} = 6$)
40×10	400	50	0.414	optimizer not converged
80×20	1600	25	0.058	Primary result
160×40	6400	12.5	0.091	600 iter, identical; J monotone

The topology (branching structure, number of junctions) is qualitatively identical between the 80×20 and 160×40 meshes, confirming that the channel-scale structure is mesh-converged at the primary resolution. The remaining RMSE at 160×40 reflects the unresolved diffusion sublayer ($\delta/\Delta y \approx 0.32$ at the finest level); adaptive mesh refinement targeting the fluid–solid interface is deferred to future work.

For larger meshes, the direct MUMPS solver is replaced by a block-triangular field-split preconditioner: algebraic multigrid (HYPRE) for the velocity block and the Schur complement approximation for the pressure [12]. Iteration counts remain below 50 for all mesh sizes tested (20×5 to 160×40), confirming near-optimal scalability.

S2: Filter Radius Sensitivity

The Helmholtz filter radius $r_f = 2\Delta x$ was varied between $1\Delta x$ and $4\Delta x$ on the 80×20 mesh. The outlet RMSE varied by less than 0.02 across this range, and the qualitative topology was preserved throughout. Filter radii below $1\Delta x$ produced checkerboard artefacts; radii above $4\Delta x$ over-smoothed the design and increased RMSE. The value $r_f = 2\Delta x$ lies in the centre of the stable range and is used throughout the main paper.

S3: SUPG Stabilisation Validation

Three forward solves on the 80×20 mesh were compared for the linear gradient target: (i) no stabilisation, (ii) SUPG with τ from Eq. (28), and (iii) full upwinding ($\tau = h/(2\|\mathbf{u}\|)$). Without stabilisation, the concentration field exhibits oscillations with amplitude up to 0.4 at the inlet interface. SUPG reduces oscillation amplitude below 10^{-3} while maintaining the smooth outlet gradient profile. Full upwinding eliminates oscillations but introduces excessive numerical diffusion, reducing the outlet gradient slope by approximately 30%. SUPG is therefore the appropriate choice for this operating regime.

S4: OC vs. MMA Convergence Comparison

Both the Optimality Criteria (OC) method and the Method of Moving The hybrid OC/MMA schedule was run for 600 iterations on the linear gradient target (80×20 mesh, identical initialisation and continuation schedules). The hybrid OC/MMA strategy (OC for topology formation at $\beta = 1$–2, MMA for sharpening at $\beta = 4$–16) converges to $J = 1.76 \times 10^{-3}$ with outlet RMSE = 0.058 in 1400 iterations.

Pure OC was tested independently and exhibits a period-2 oscillation at $\beta \geq 4$: the objective alternates between $J \approx 3.6 \times 10^{-3}$ and $J \approx 5.9 \times 10^{-3}$ on every iteration, converging to neither value. Diagnosis revealed that the sensitivity vector has a near-symmetric sign distribution (47% negative, 53% positive), causing the OC bisection to find two complementary designs that it alternates between indefinitely. MMA's asymptote-based move control breaks this cycle by adaptively contracting asymptotes in oscillating directions.

The self-contained Svanberg dual-bisection MMA implementation (`micrograd/optimizer.py`) requires no external packages. The volume constraint is enforced explicitly via dual bisection in the MMA subproblem, achieving $|V - V^*| < 10^{-3}$ at every iteration once the topology is established.

S10: Adjoint Gradient Verification Details

Two verification tests are reported in Section 4.1.

FD Taylor remainder. The finite-difference directional derivative $\hat{g}[\delta\rho] = (J(\rho + \varepsilon\delta\rho) - J(\rho))/\varepsilon$ at $\varepsilon = 10^{-5}$ is used as the reference. The Taylor remainder $R(\varepsilon) = |J(\rho + \varepsilon\delta\rho) - J(\rho) - \varepsilon\hat{g}[\delta\rho]|$ is computed for $\varepsilon \in \{10^{-5}, \ldots, 10^{-2}\}$. Least-squares slopes: 2.12 (20×5), 2.07 (80×20).

Direction test. The mass-scaled adjoint g/m_i is compared to the finite-difference gradient $\hat{g}_i$ component-wise on 20 sampled DOFs. Pearson correlations: 0.80 (20×5), 0.88 (80×20). The magnitude discrepancy reflects the difference between lumped and full mass-matrix Riesz maps; only the direction is used by OC/MMA.

Continuous vs. discrete adjoint. The assembled sensitivity is a continuous adjoint $g_i = \int(\partial J/\partial\bar{\rho})\psi_i\,dx$, not the discrete gradient $\hat{g}_i = \partial J/\partial\bar{\rho}_i$. The relative error $E_g = \|g/m - \hat{g}\|/\|\hat{g}\| = 18.9$ (20×5) reflects this mismatch. A discrete adjoint is deferred to future work.

S5: Thresholding Robustness and Binary Gap Analysis

The optimised design has a grey-zone fraction of 0.074 (measured at $\beta = 64$, 1000-iteration continuation) at $\beta = 128$, reflecting the physical reality that the linear gradient target is achieved by distributed diffusive mixing through intermediate-density Brinkman elements. Threshold-projection at $\bar{\rho} = 0.5$ and re-solving the forward problem gives the results in Table 7.

Table 7: Binary validation results on the 80×20 mesh. Both binary configurations give identical RMSE, confirming the gap is caused by loss of gray mixing pathways, not by the Brinkman penalisation in fluid regions.

Design	**RMSE**	**ΔRMSE vs grey**	**Method**
Grey Brinkman	0.058	—	Smooth $\alpha(\bar{\rho})$
Binary Brinkman	0.359	+356%	$\bar{\rho} \in \{0, 1\}$, smooth α

The + + 356% RMSE change is mechanistically expected: the linear gradient target $c^\star(y) = y/L_y$ is optimally achieved by diffusive transport through intermediate-density elements, which have no binary physical analogue. The gap is not caused by the Brinkman penalisation in fluid regions but by the loss of the gray diffusive mixing pathway. The analytical bound $t_{\text{diff}}/t_{\text{flow}} \approx 3000$ (Section 2.3) confirms that solid leakage through $D_{\min}$ is negligible; the binary gap is a property of the topology, not a numerical artefact. Body-fitted Navier–Stokes validation on a purpose-designed binary multi-channel manifold is deferred to future work.

References

[1] T. Borrvall and J. Petersson. Topology optimization of fluids in stokes flow. *International Journal for Numerical Methods in Fluids*, 41(1):77–107, 2003.

[2] J. Alexandersen, C. S. Andreasen, N. Aage, and O. Sigmund. Topology optimisation for natural convection problems. *International Journal for Numerical Methods in Fluids*, 76(10):699–721, 2014.

[3] D. S. Makhija and K. Maute. Level set topology optimization of scalar transport problems. *Structural and Multidisciplinary Optimization*, 51:267–285, 2015.

[4] S. Kreissl, G. Pingen, and K. Maute. Topology optimization for unsteady flow. *International Journal for Numerical Methods in Engineering*, 87(13):1229–1253, 2011.

[5] George M. Whitesides. The origins and the future of microfluidics. *Nature*, 442:368–373, 2006.

[6] J. Na, H. Li, P. Yan, X. Li, and X. Gao. An open-source topology optimization modeling framework for the design of passive micromixer structure. *Chemical Engineering Science*, 260:117820, 2022.

[7] Jørgen S. Dokken, August Johansson, Mikael Ranta, and Matthew W. Scroggs. Automatic finite element code generation for variational problems in FEniCSx. *ACM Transactions on Mathematical Software*, 49(4):1–26, 2023.

[8] Boyan S. Lazarov and Ole Sigmund. Filters in topology optimization based on Helmholtz-type differential equations. *International Journal for Numerical Methods in Engineering*, 86(6):765–781, 2011.

[9] A. N. Brooks and T. J. R. Hughes. Streamline upwind/petrov-galerkin formulations for convection dominated flows with particular emphasis on the incompressible navier-stokes equations. *Computer Methods in Applied Mechanics and Engineering*, 32:199–259, 1982.

[10] F. Wang, B. S. Lazarov, and O. Sigmund. On projection methods, convergence and robust formulations in topology optimization. *Structural and Multidisciplinary Optimization*, 43:767–784, 2011.

[11] Igor A. Baratta, Joseph P. Dean, Jørgen S. Dokken, Michal Habera, Jack S. Hale, Chris N. Richardson, Marie E. Ringøse, Marie E. Rognes, Matthew W. Scroggs, Nathan Sime, and Garth N. Wells. DOLFINx: The next generation FEniCS problem solving environment. *preprint*, 2023.

[12] Howard C. Elman, David J. Silvester, and Andrew J. Wathen. *Finite Elements and Fast Iterative Solvers: With Applications in Incompressible Fluid Dynamics*. Oxford University Press, 2014.

[13] M. P. Bendsøe and O. Sigmund. *Topology Optimization – Theory, Methods, and Applications*. Springer, 2003.

[14] K. Svanberg. The method of moving asymptotes – a new method for structural optimization. *International Journal for Numerical Methods in Engineering*, 24:359–373, 1987.

[15] Nisong Monyimba. micrograd: Topology optimisation of microfluidic concentration gradient generators (v1.0.0). `https://github.com/nisongmonyimba278-byte/micrograd`, 2025. Zenodo DOI pending registration; code available at GitHub.

[16] GitHub, Inc. GitHub Actions: Automate your workflow. `https://github.com/features/actions`, 2024. Accessed: 2025.